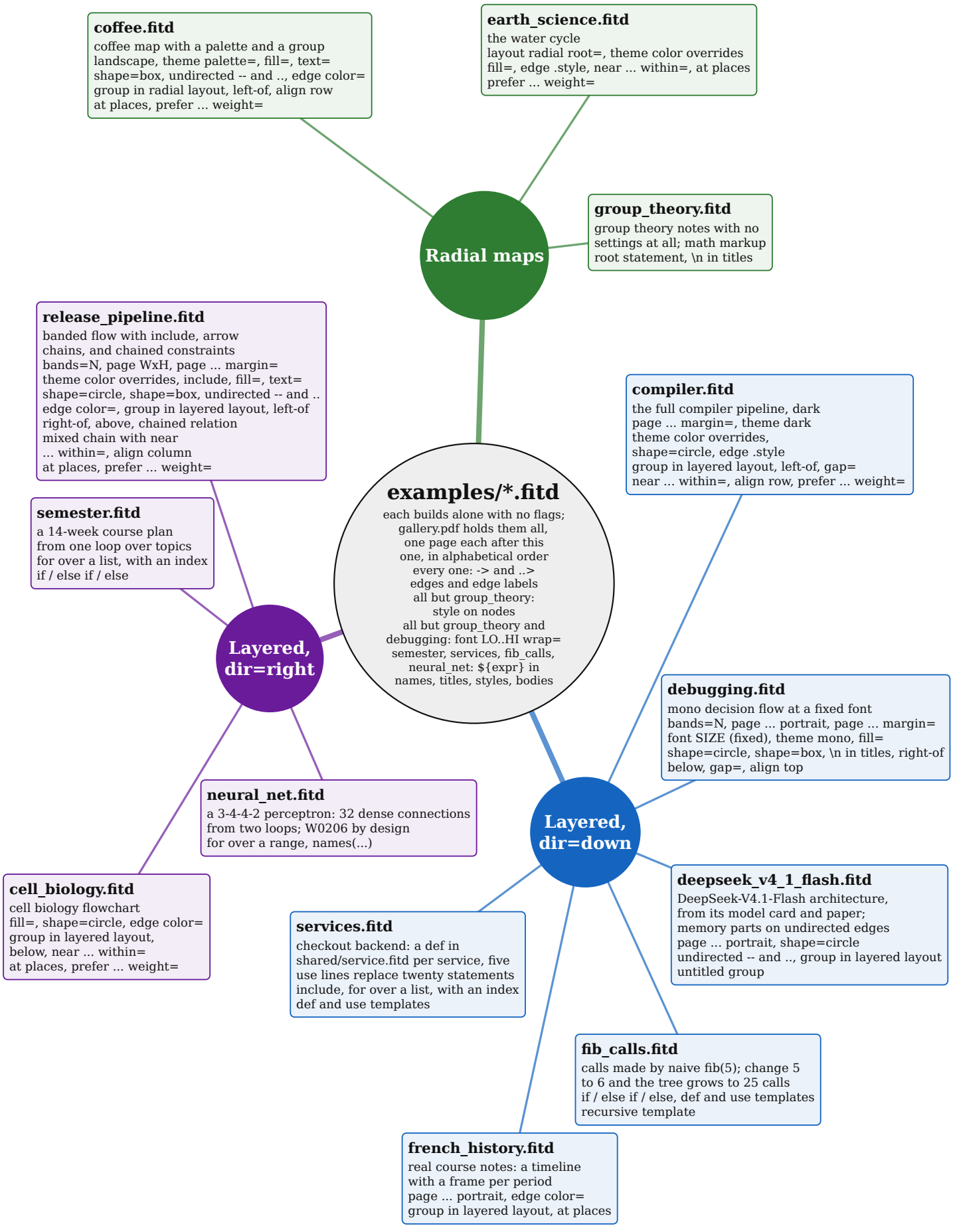
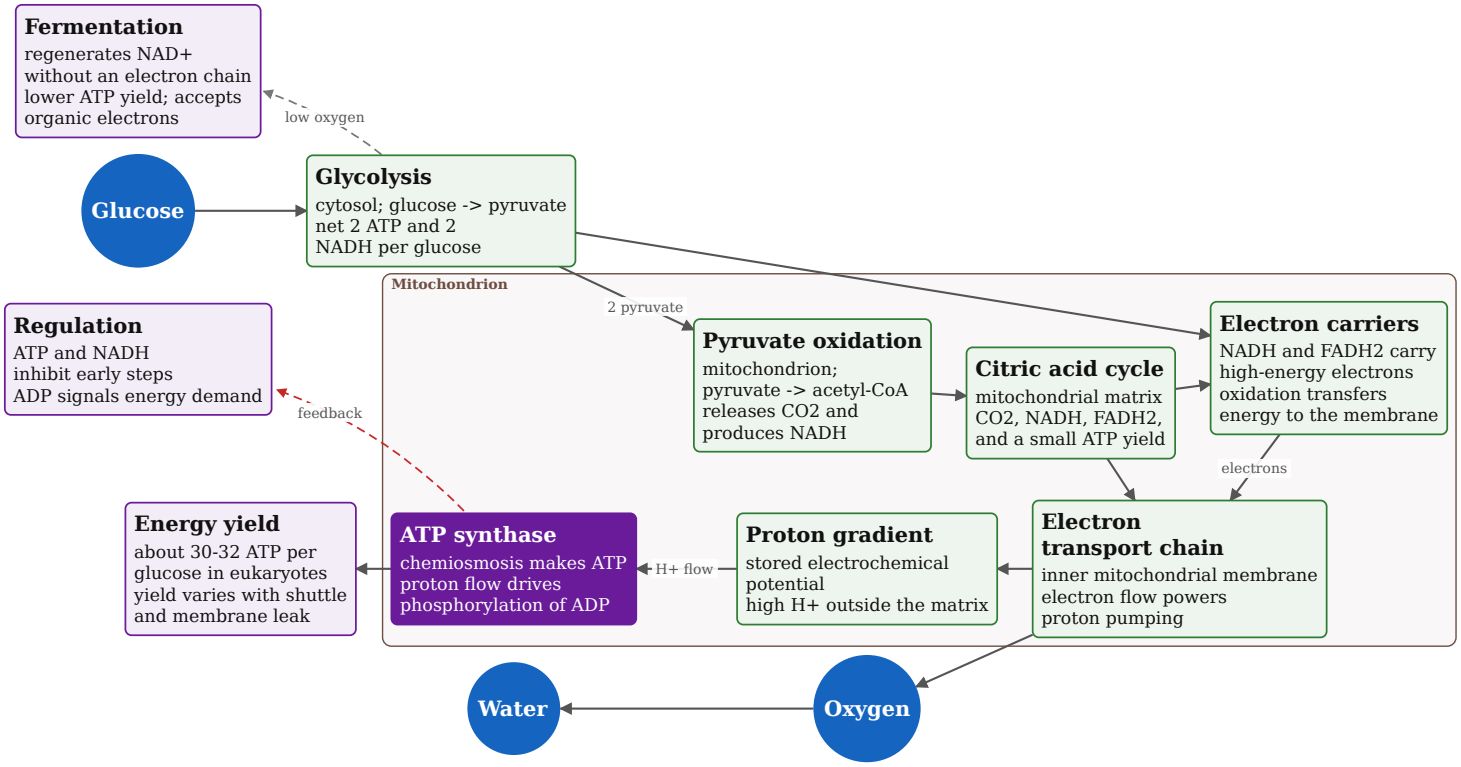
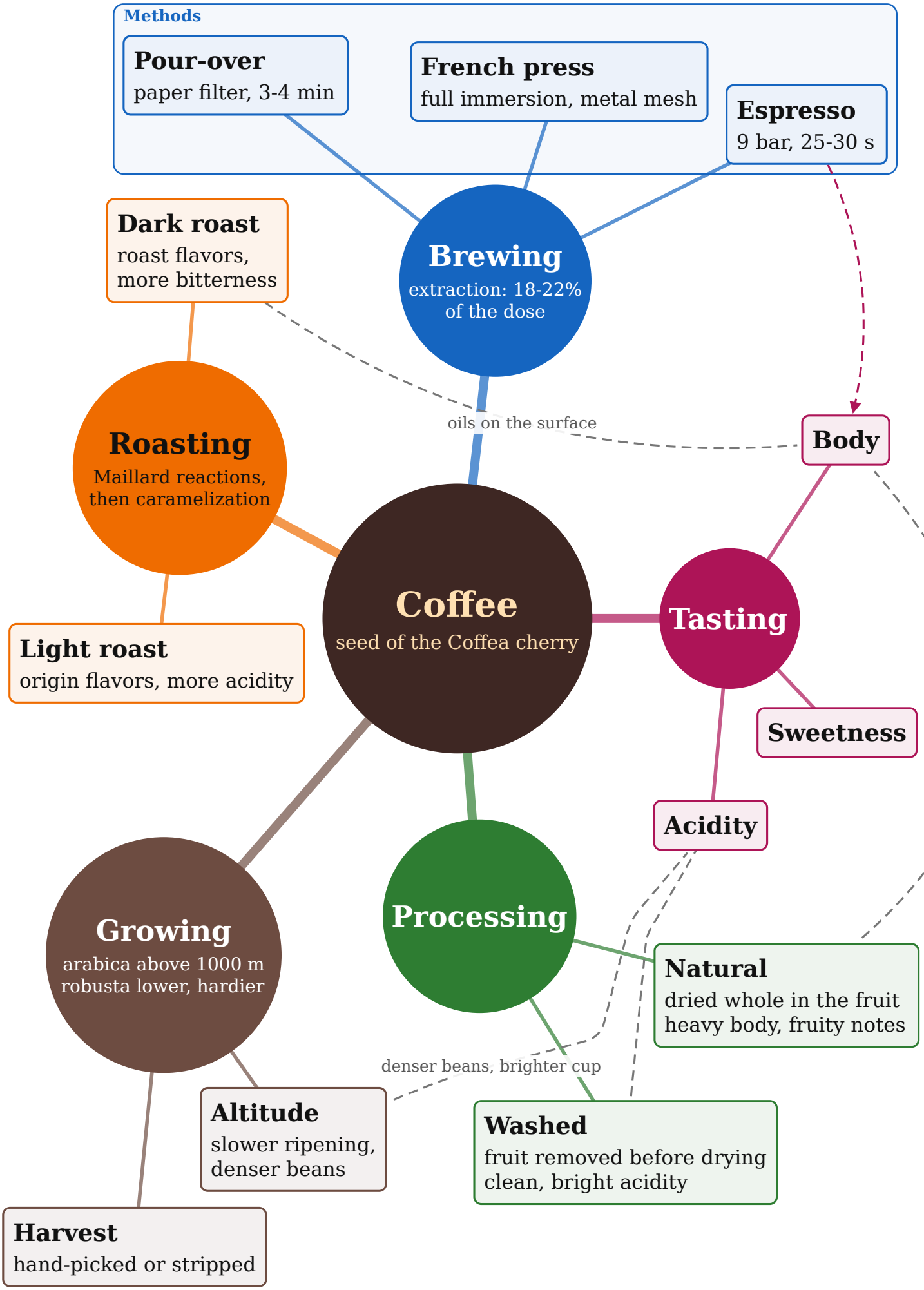
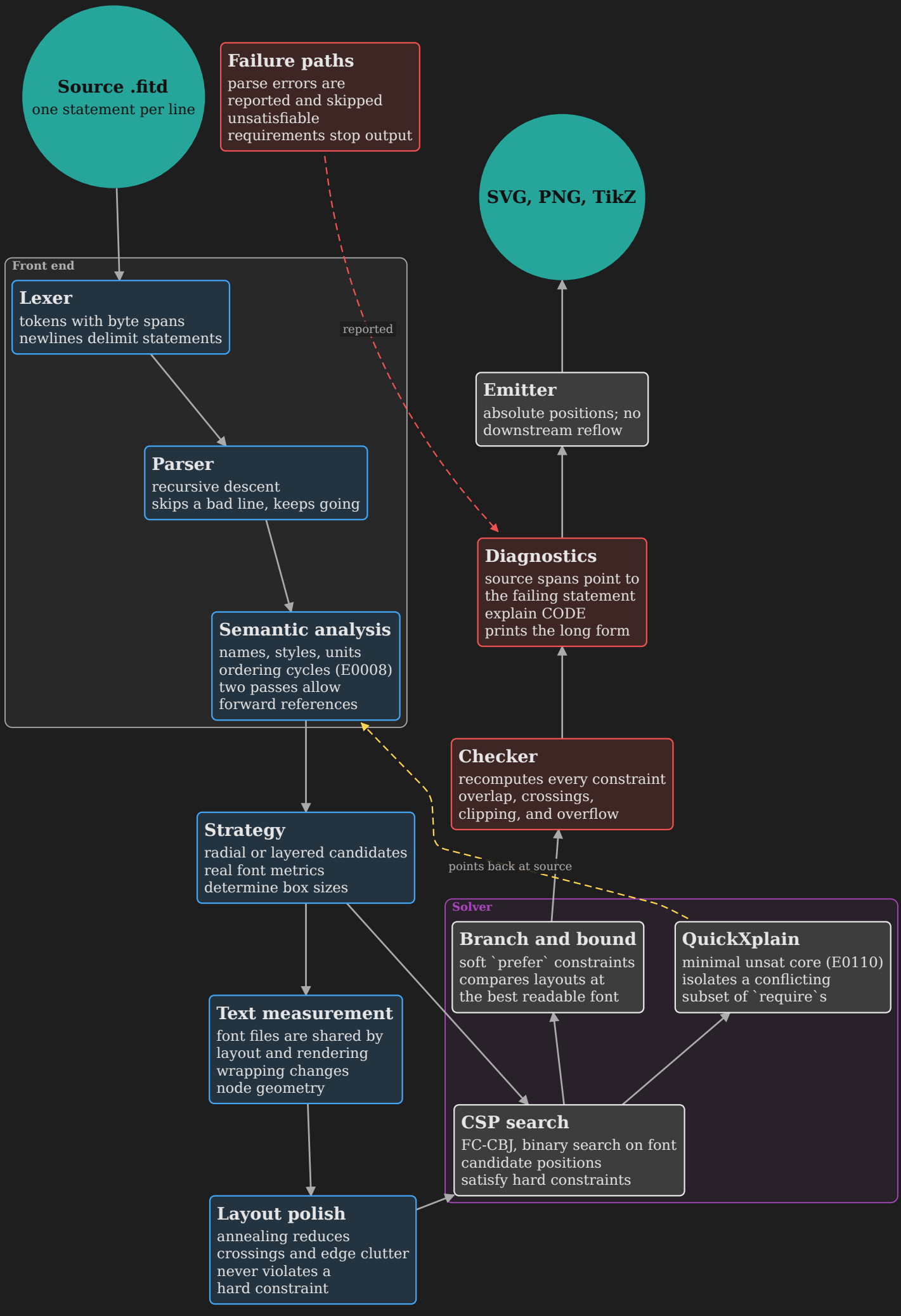
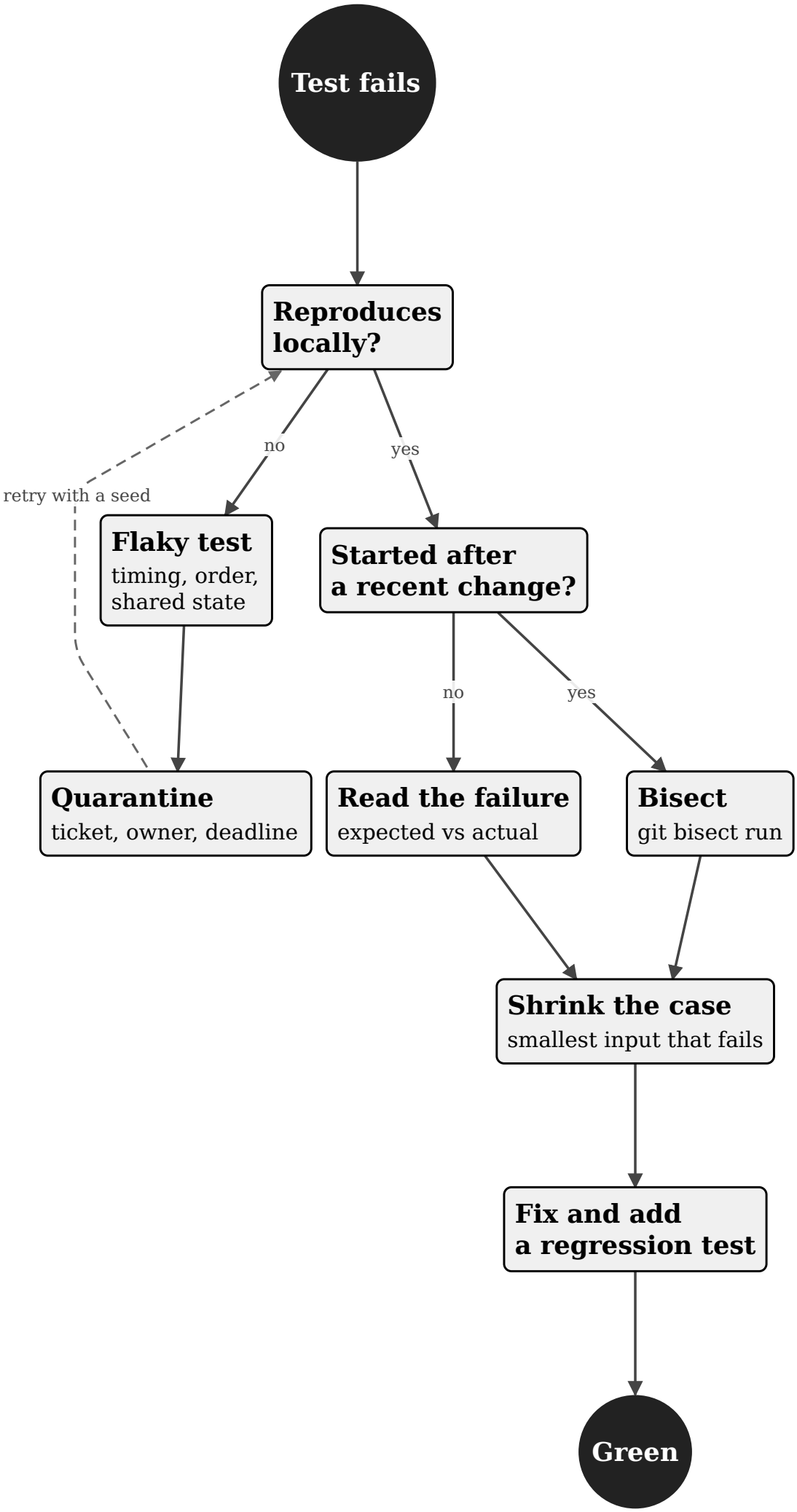
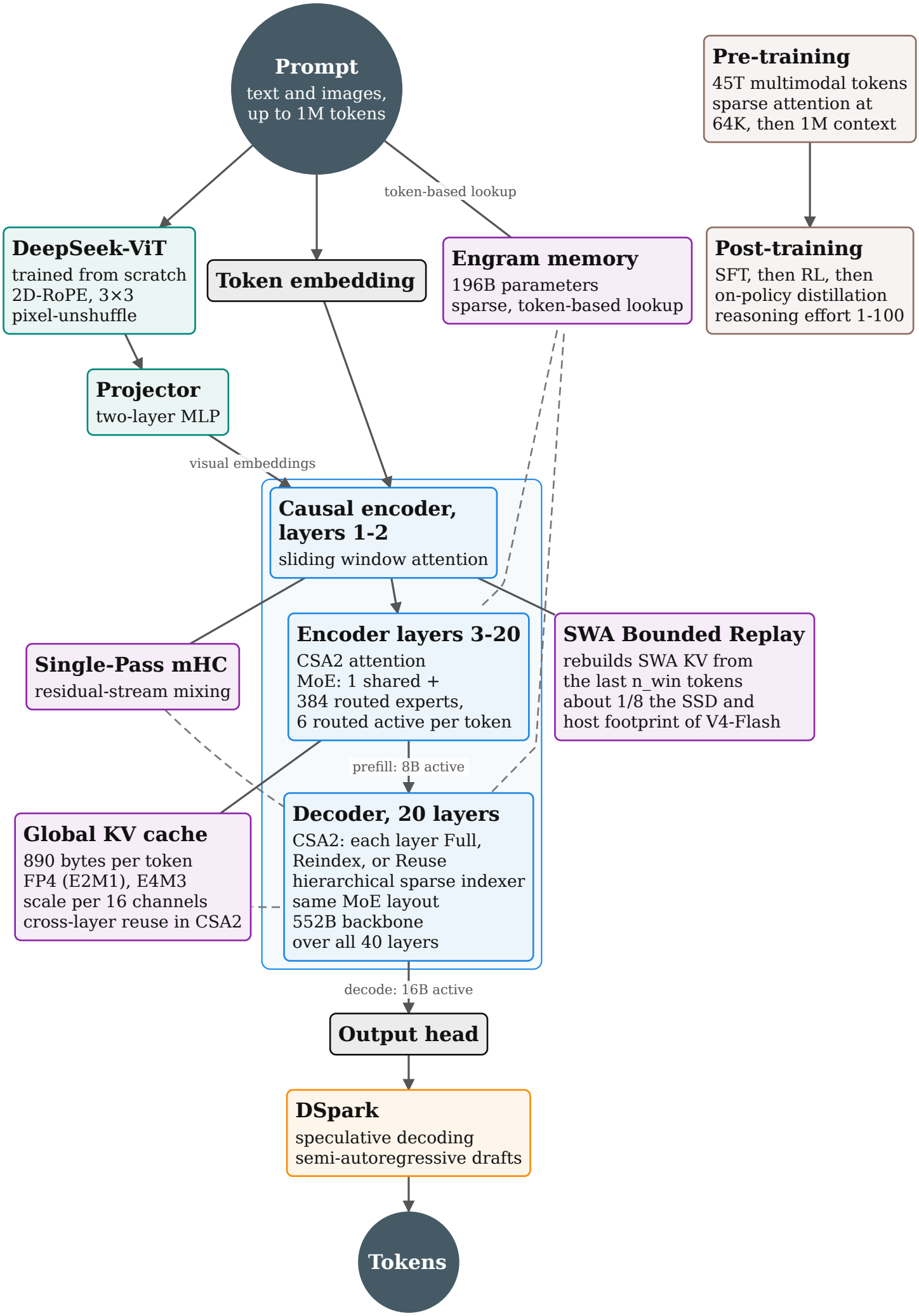

examples/*.fitd
each builds alone with no flags; gallery.pdf holds them all, one page each after this one, in alphabetical order
every one: -> and ..>
edges and edge labels
all but group_theory: style on nodes
all but group_theory and debugging: font LO..HI wrap= semester, services, fib_calls, neural_net: ${expr} in names, titles, styles, bodies
Radial maps
coffee.fitd
coffee map with a palette and a group
landscape, theme palette=, fill=, text=
shape=box, undirected -- and .., edge color=
group in radial layout, left-of, align row
at places, prefer ... weight=
earth_science.fitd
the water cycle
layout radial root=, theme color overrides
fill=, edge .style, near ... within=, at places
prefer ... weight=
group_theory.fitd
group theory notes with no
settings at all; math markup
root statement, \n in titles
Layered, dir=right
release_pipeline.fitd
banded flow with include, arrow
chains, and chained constraints
bands=N, page WxH, page ... margin=
theme color overrides, include, fill=, text=
shape=circle, shape=box, undirected -- and ..
edge color=, group in layered layout, left-of
right-of, above, chained relation
mixed chain with near
... within=, align column
at places, prefer ... weight=
semester.fitd
a 14-week course plan
from one loop over topics
for over a list, with an index
if / else if / else
neural_net.fitd
a 3-4-4-2 perceptron: 32 dense connections
from two loops; W0206 by design
for over a range, names(...)
cell_biology.fitd
cell biology flowchart
fill=, shape=circle, edge color=
group in layered layout,
below, near ... within=
at places, prefer ... weight=
Layered, dir=down
compiler.fitd
the full compiler pipeline, dark
page ... margin=, theme dark
theme color overrides,
shape=circle, edge .style
group in layered layout, left-of, gap=
near ... within=, align row, prefer ... weight=
debugging.fitd
mono decision flow at a fixed font
bands=N, page ... portrait, page ... margin=
font SIZE (fixed), theme mono, fill=
shape=circle, shape=box, \n in titles, right-of
below, gap=, align top
deepseek_v4_1_flash.fitd
DeepSeek-V4.1-Flash architecture,
from its model card and paper;
memory parts on undirected edges
page ... portrait, shape=circle
undirected -- and .., group in layered layout
untitled group
services.fitd
checkout backend: a def in
shared/service.fitd per service, five
use lines replace twenty statements
include, for over a list, with an index
def and use templates
fib_calls.fitd
calls made by naive fib(5); change 5
to 6 and the tree grows to 25 calls
if / else if / else, def and use templates
recursive template
french_history.fitd
real course notes: a timeline
with a frame per period
page ... portrait, edge color=
group in layered layout, at places

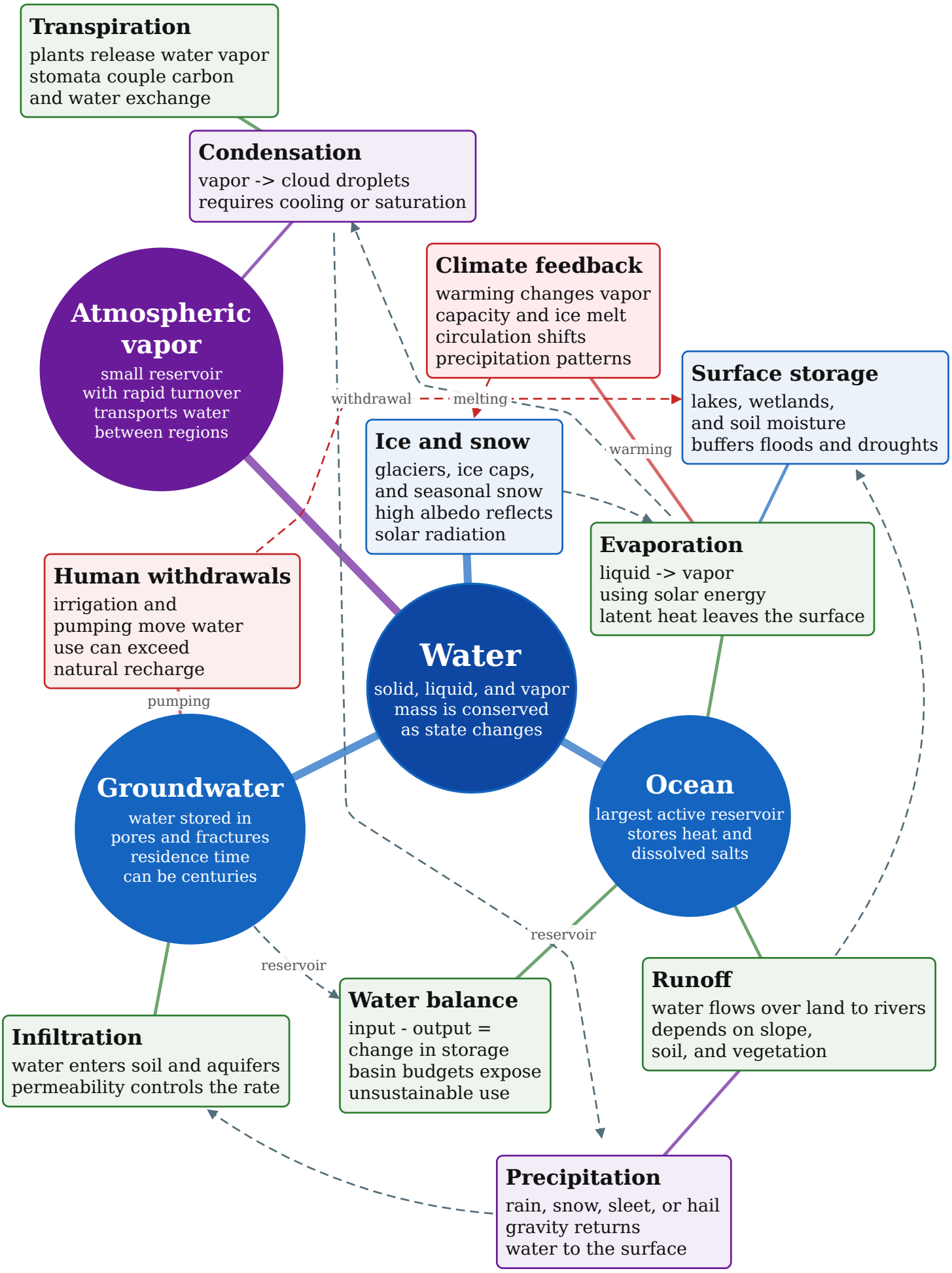

Fermentation
regenerates NAD+
without an electron chain
lower ATP yield; accepts
organic electrons
low oxygen
Glucose
Glycolysis
cytosol; glucose -> pyruvate
net 2 ATP and 2
NADH per glucose
Mitochondrion
2 pyruvate
Pyruvate oxidation
mitochondrion;
pyruvate -> acetyl-CoA
releases CO2 and
produces NADH
Citric acid cycle
mitochondrial matrix
CO2, NADH, FADH2,
and a small ATP yield
Electron carriers
NADH and FADH2 carry
high-energy electrons
oxidation transfers
energy to the membrane
electrons
Regulation
ATP and NADH
inhibit early steps
ADP signals energy demand
feedback
Energy yield
about 30-32 ATP per
glucose in eukaryotes
yield varies with shuttle
and membrane leak
ATP synthase
chemiosmosis makes ATP
proton flow drives
phosphorylation of ADP
H+ flow
Proton gradient
stored electrochemical
potential
high H+ outside the matrix
Electron
transport chain
inner mitochondrial membrane
electron flow powers
proton pumping
Water
Oxygen

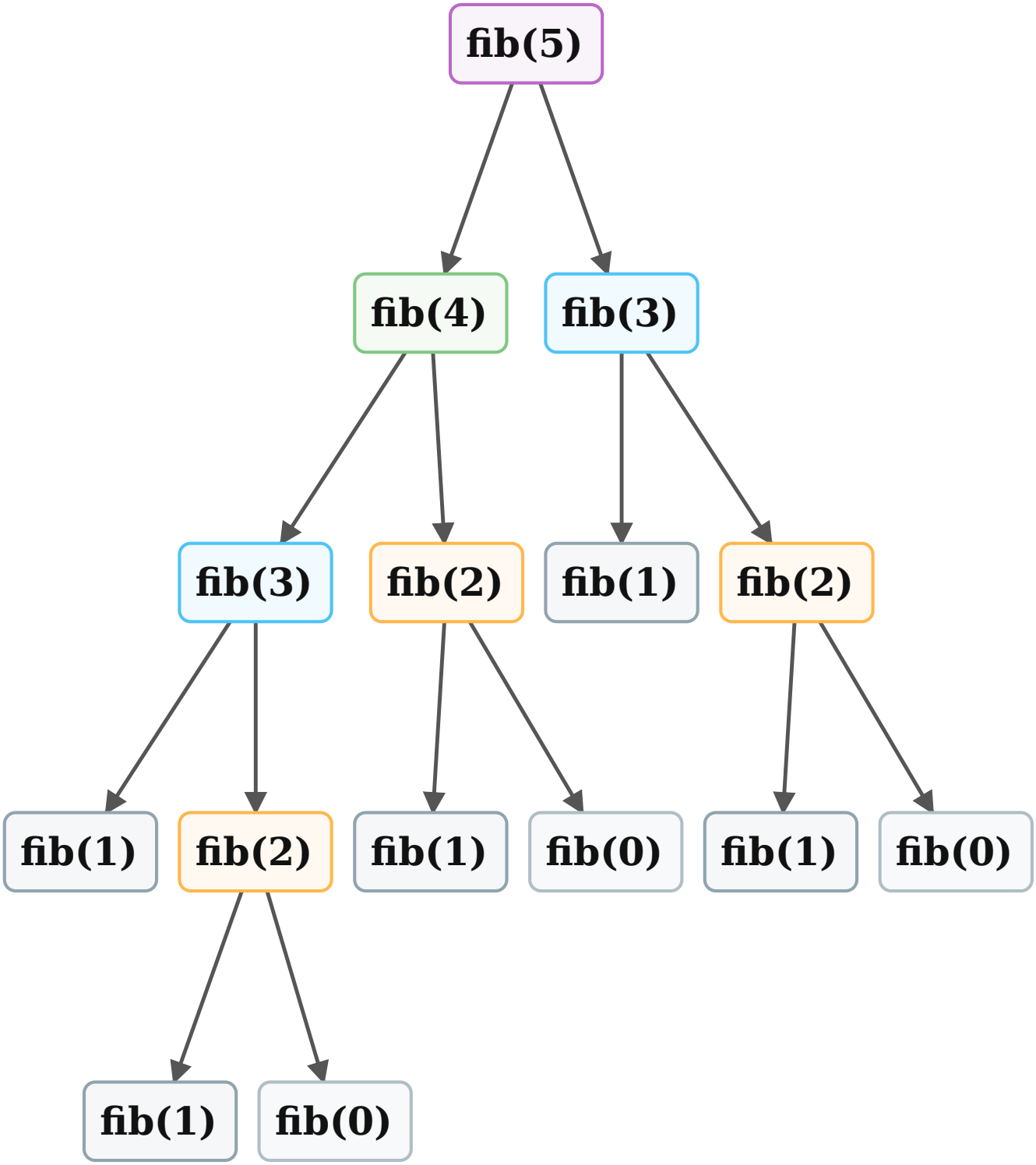

Methods
Pour-over
paper filter, 3-4 min
French press
full immersion, metal mesh
Espresso
9 bar, 25-30 s
Brewing
extraction: 18-22% of the dose
Dark roast
roast flavors, more bitterness
Roasting
Maillard reactions, then caramelization
Light roast
origin flavors, more acidity
oils on the surface
Body
Coffee
seed of the Coffea cherry
Tasting
Sweetness
Acidity
Growing
arabica above 1000 m
robusta lower, hardier
Processing
Natural
dried whole in the fruit
heavy body, fruity notes
denser beans, brighter cup
Altitude
slower ripening, denser beans
Washed
fruit removed before drying
clean, bright acidity
Harvest
hand-picked or stripped

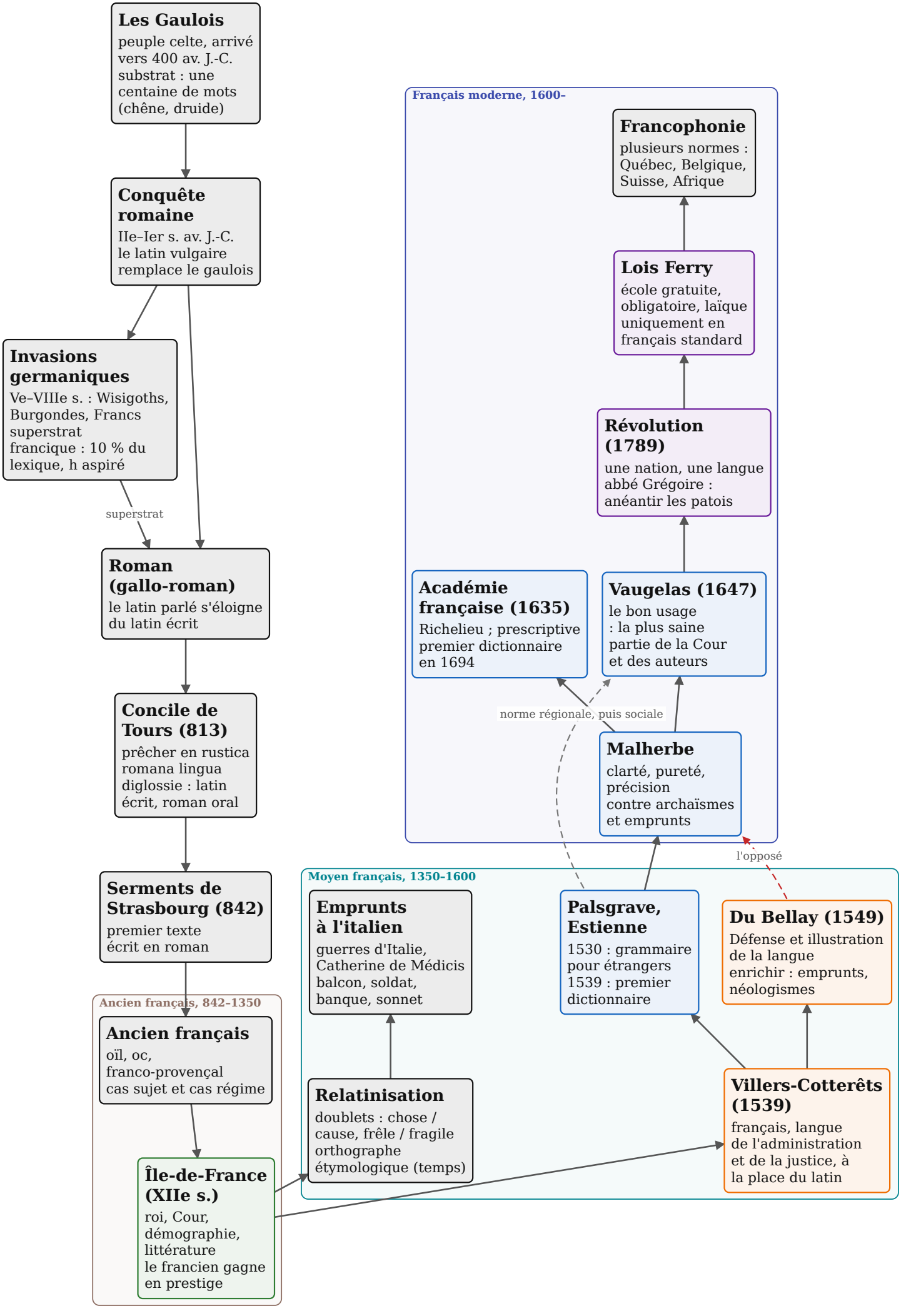

Source .fitd
one statement per line
Failure paths
parse errors are reported and skipped
unsatisfiable requirements stop output
SVG, PNG, TikZ
Front end
Lexer
tokens with byte spans
newlines delimit statements
reported
Emitter
absolute positions; no downstream reflow
Parser
recursive descent
skips a bad line, keeps going
Diagnostics
source spans point to the failing statement
explain CODE
prints the long form
Semantic analysis
names, styles, units
ordering cycles (E0008)
two passes allow forward references
Checker
recomputes every constraint
overlap, crossings, clipping, and overflow
Strategy
radial or layered candidates
real font metrics
determine box sizes
points back at source
Solver
Branch and bound
soft `prefer` constraints
compares layouts at the best readable font
QuickXplain
minimal unsat core (E0110)
isolates a conflicting subset of `require`s
Text measurement
font files are shared by layout and rendering
wrapping changes node geometry
CSP search
FC-CBJ, binary search on font candidate positions
satisfy hard constraints
Layout polish
annealing reduces crossings and edge clutter
never violates a hard constraint

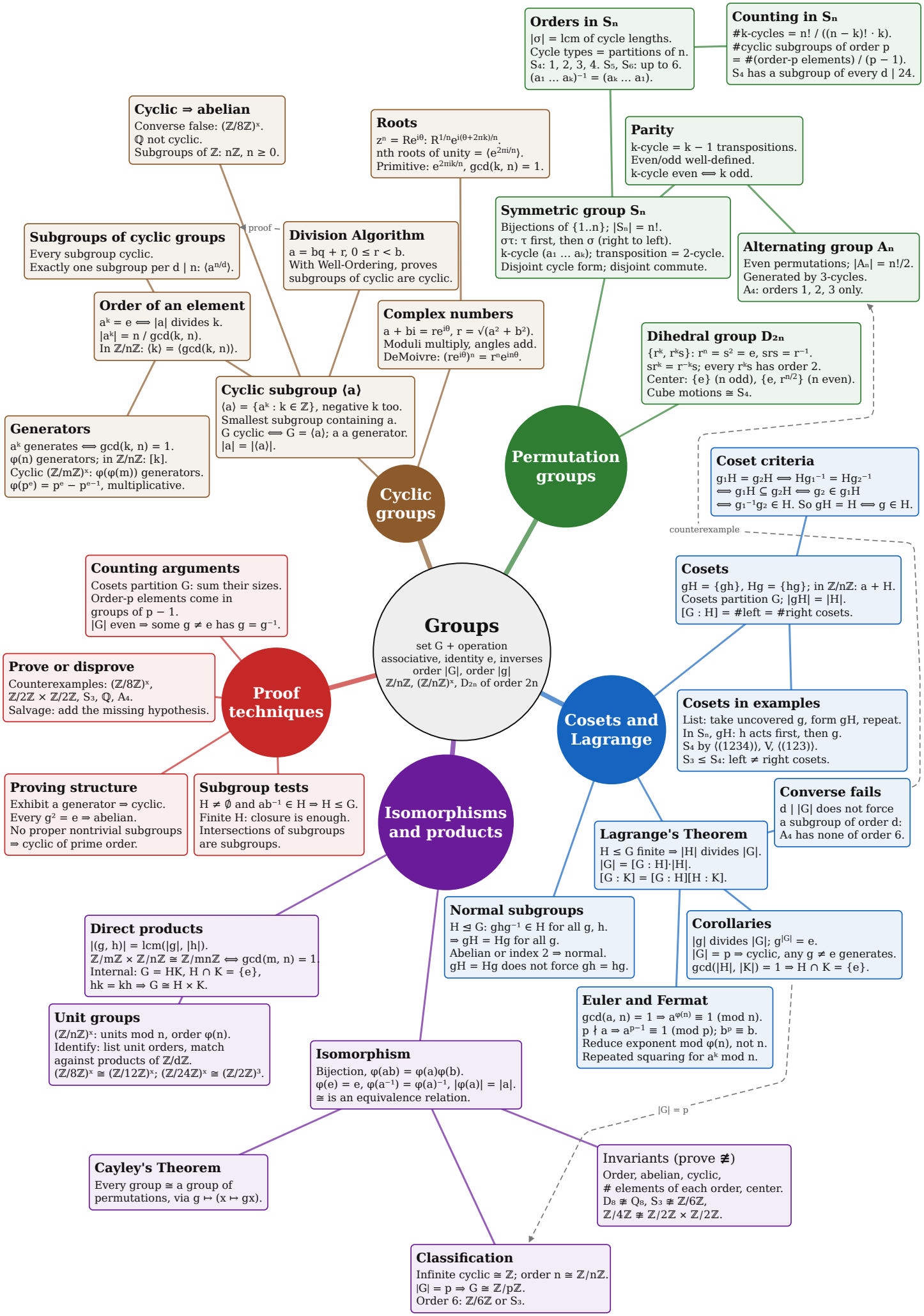

Test fails
Reproduces locally?
no
yes
retry with a seed
Flaky test
timing, order, shared state
Started after a recent change?
no
yes
Quarantine
ticket, owner, deadline
Read the failure
expected vs actual
Bisect
git bisect run
Shrink the case
smallest input that fails
Fix and add a regression test
Green

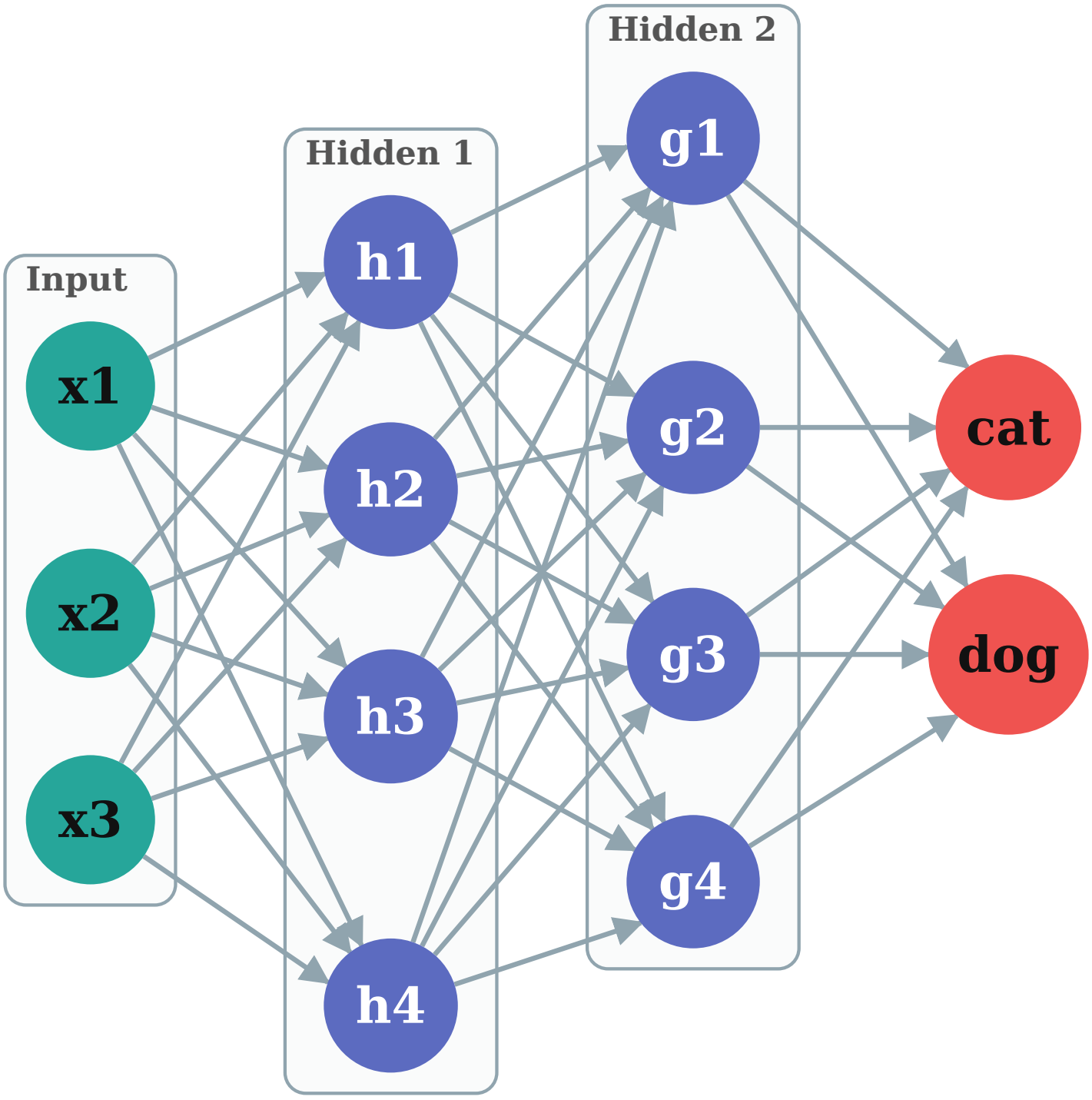

Prompt
text and images, up to 1M tokens
Pre-training
45T multimodal tokens
sparse attention at 64K, then 1M context
token-based lookup
DeepSeek-ViT
trained from scratch
2D-RoPE, 3×3 pixel-unshuffle
Token embedding
Engram memory
196B parameters
sparse, token-based lookup
Post-training
SFT, then RL, then on-policy distillation
reasoning effort 1-100
Projector
two-layer MLP
visual embeddings
Causal encoder, layers 1-2
sliding window attention
Encoder layers 3-20
CSA2 attention
MoE: 1 shared + 384 routed experts, 6 routed active per token
SWA Bounded Replay
rebuilds SWA KV from the last n_win tokens
about 1/8 the SSD and host footprint of V4-Flash
Single-Pass mHC
residual-stream mixing
prefill: 8B active
Global KV cache
890 bytes per token
FP4 (E2M1), E4M3 scale per 16 channels
cross-layer reuse in CSA2
Decoder, 20 layers
CSA2: each layer Full, Reindex, or Reuse
hierarchical sparse indexer
same MoE layout
552B backbone
over all 40 layers
decode: 16B active
Output head
DSpark
speculative decoding
semi-autoregressive drafts
Tokens

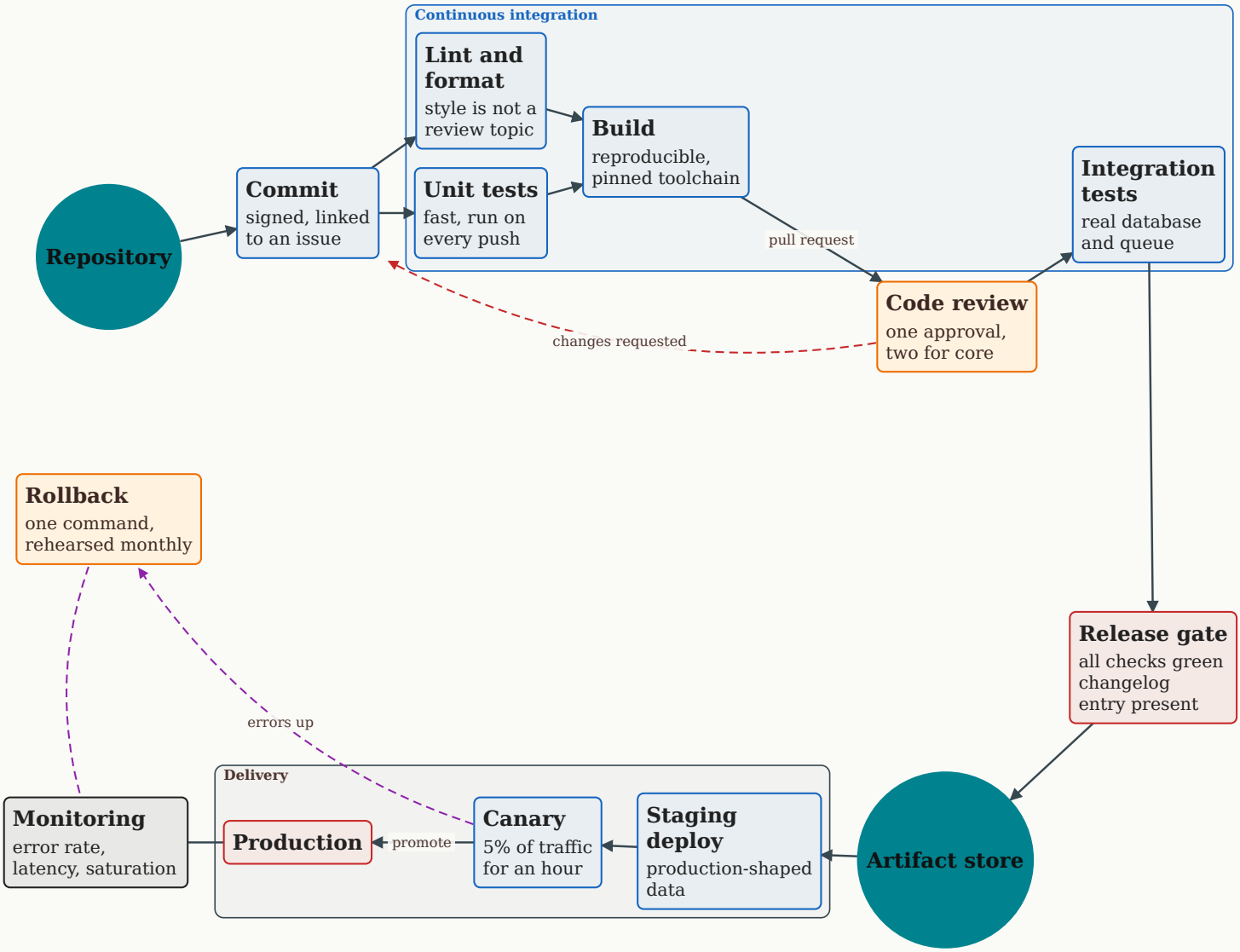

Transpiration
plants release water vapor
stomata couple carbon
and water exchange
Condensation
vapor -> cloud droplets
requires cooling or saturation
Atmospheric vapor
small reservoir
with rapid turnover
transports water
between regions
Climate feedback
warming changes vapor
capacity and ice melt
circulation shifts
precipitation patterns
withdrawal
melting
warming
Surface storage
lakes, wetlands,
and soil moisture
buffers floods and droughts
Ice and snow
glaciers, ice caps,
and seasonal snow
high albedo reflects
solar radiation
Evaporation
liquid -> vapor
using solar energy
latent heat leaves the surface
Human withdrawals
irrigation and
pumping move water
use can exceed
natural recharge
Water
solid, liquid, and vapor
mass is conserved
as state changes
pumping
Groundwater
water stored in
pores and fractures
residence time
can be centuries
Ocean
largest active reservoir
stores heat and
dissolved salts
reservoir
reservoir
Runoff
water flows over land to rivers
depends on slope,
soil, and vegetation
Water balance
input - output =
change in storage
basin budgets expose
unsustainable use
Infiltration
water enters soil and aquifers
permeability controls the rate
Precipitation
rain, snow, sleet, or hail
gravity returns
water to the surface

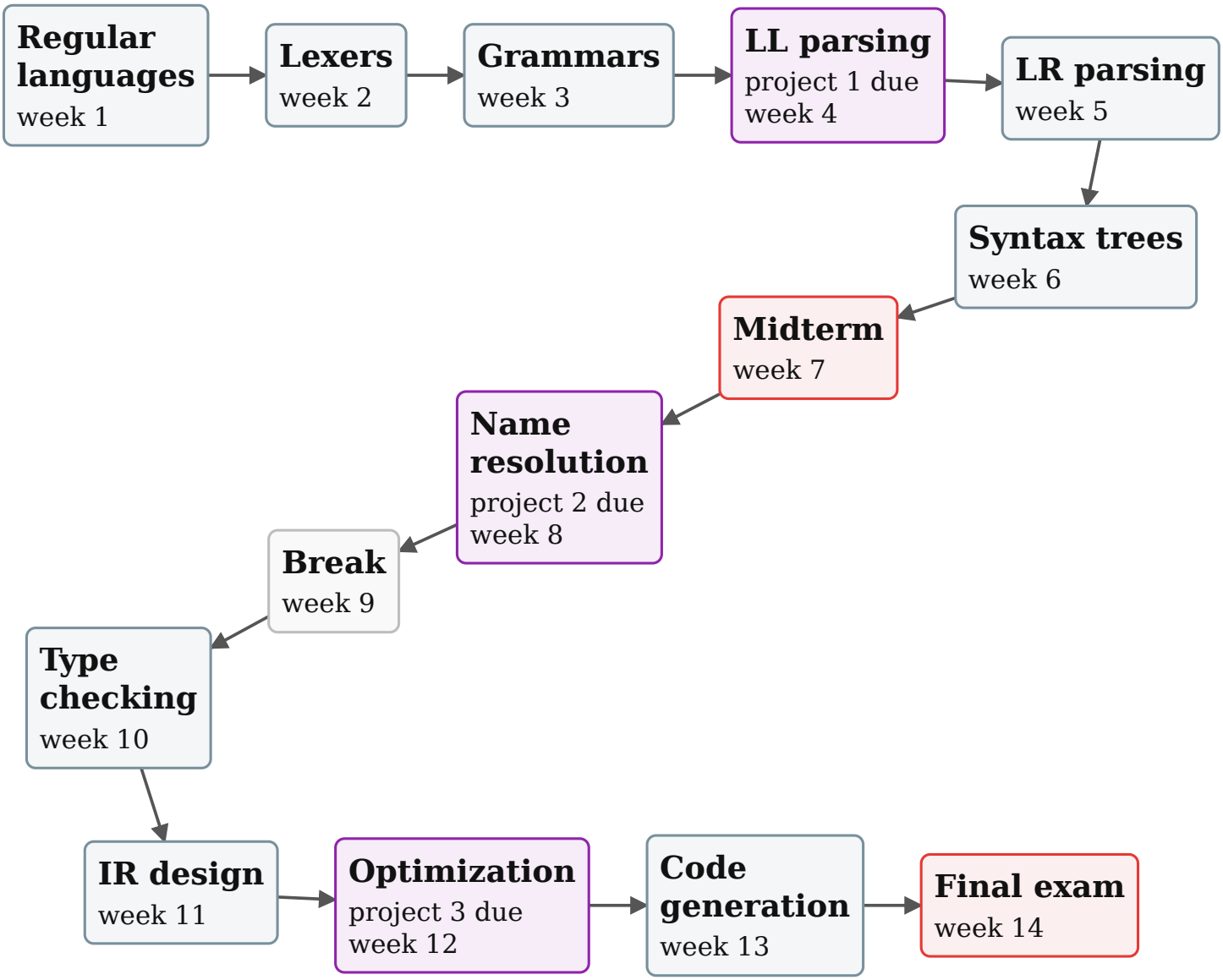

fib(5)
fib(4)
fib(3)
fib(3)
fib(2)
fib(1)
fib(2)
fib(1)
fib(2)
fib(1)
fib(0)
fib(1)
fib(0)
fib(1)
fib(0)

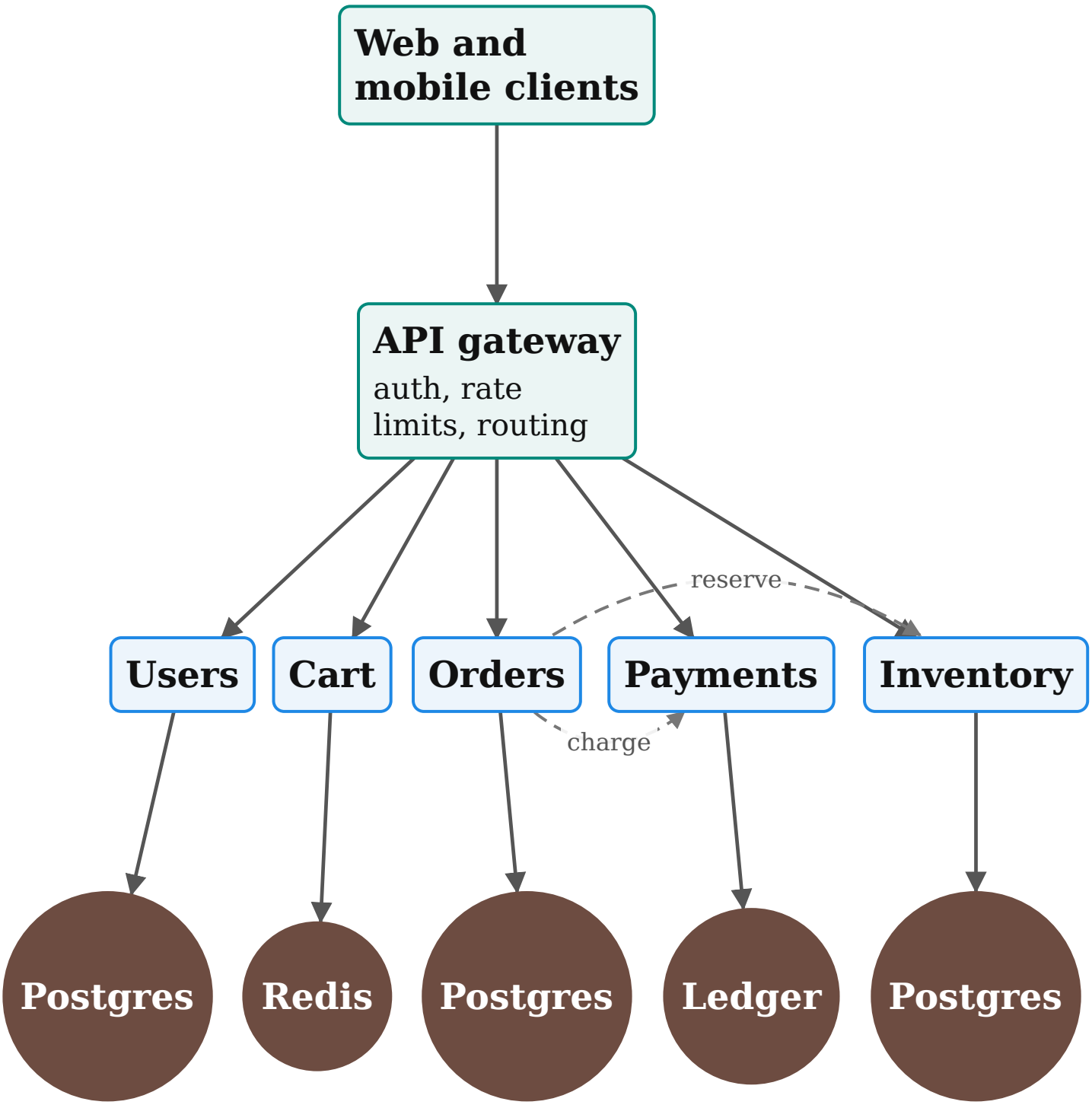

Les Gaulois
peuple celte, arrivé vers 400 av. J.-C.
substrat : une centaine de mots (chêne, druide)
Conquête romaine
IIe–Ier s. av. J.-C.
le latin vulgaire remplace le gaulois
Invasions germaniques
Ve–VIIIe s. : Wisigoths, Burgondes, Francs
superstrat francique : 10 % du lexique, h aspiré
superstrat
Roman (gallo-roman)
le latin parlé s'éloigne du latin écrit
Concile de Tours (813)
prêcher en rustica romana lingua
diglossie : latin écrit, roman oral
Serments de Strasbourg (842)
premier texte écrit en roman
Ancien français, 842–1350
Ancien français
oïl, oc, franco-provençal
cas sujet et cas régime
Île-de-France (XIIe s.)
roi, Cour, démographie, littérature
le francien gagne en prestige
Français moderne, 1600–
Francophonie
plusieurs normes : Québec, Belgique, Suisse, Afrique
Lois Ferry
école gratuite, obligatoire, laïque
uniquement en français standard
Révolution (1789)
une nation, une langue
abbé Grégoire : anéantir les patois
Académie française (1635)
Richelieu ; prescriptive
premier dictionnaire en 1694
Vaugelas (1647)
le bon usage : la plus saine partie de la Cour et des auteurs
norme régionale, puis sociale
Malherbe
clarté, pureté, précision
contre archaïsmes et emprunts
l'opposé
Moyen français, 1350–1600
Emprunts à l'italien
guerres d'Italie, Catherine de Médicis
balcon, soldat, banque, sonnet
Palsgrave, Estienne
1530 : grammaire pour étrangers
1539 : premier dictionnaire
Du Bellay (1549)
Défense et illustration de la langue
enrichir : emprunts, néologismes
Relatinisation
doublets : chose / cause, frêle / fragile
orthographe étymologique (temps)
Villers-Cotterêts (1539)
français, langue de l'administration et de la justice, à la place du latin


Groups
set G + operation
associative, identity e, inverses
order |G|, order |g|
ℤ/nℤ, (ℤ/nℤ)ˣ, D₂ₙ of order 2n
Cyclic groups
Permutation groups
Proof techniques
Cosets and Lagrange
Isomorphisms and products
Orders in Sₙ
|σ| = lcm of cycle lengths.
Cycle types = partitions of n.
S₄: 1, 2, 3, 4. S₅, S₆: up to 6.
(a₁ … aₖ)⁻¹ = (aₖ … a₁).
Counting in Sₙ
#k-cycles = n! / ((n − k)! · k).
#cyclic subgroups of order p
= #(order-p elements) / (p − 1).
S₄ has a subgroup of every d | 24.
Parity
k-cycle = k − 1 transpositions.
Even/odd well-defined.
k-cycle even ⟺ k odd.
Symmetric group Sₙ
Bijections of {1..n}; |Sₙ| = n!.
στ: τ first, then σ (right to left).
k-cycle (a₁ … aₖ); transposition = 2-cycle.
Disjoint cycle form; disjoint commute.
Alternating group Aₙ
Even permutations; |Aₙ| = n!/2.
Generated by 3-cycles.
A₄: orders 1, 2, 3 only.
Dihedral group D₂ₙ
{rᵏ, rᵏs}: rⁿ = s² = e, srs = r⁻¹.
srᵏ = r⁻ᵏs; every rᵏs has order 2.
Center: {e} (n odd), {e, r^{n/2}} (n even).
Cube motions ≅ S₄.
Cyclic ⇒ abelian
Converse false: (ℤ/8ℤ)ˣ.
ℚ not cyclic.
Subgroups of ℤ: nℤ, n ≥ 0.
Roots
zⁿ = Re^{iθ}: R^{1/n}e^{i(θ+2πk)/n}.
nth roots of unity = ⟨e^{2πi/n}⟩.
Primitive: e^{2πik/n}, gcd(k, n) = 1.
proof –
Subgroups of cyclic groups
Every subgroup cyclic.
Exactly one subgroup per d | n: ⟨a^{n/d}⟩.
Division Algorithm
a = bq + r, 0 ≤ r < b.
With Well-Ordering, proves
subgroups of cyclic are cyclic.
Order of an element
aᵏ = e ⟺ |a| divides k.
|aᵏ| = n / gcd(k, n).
In ℤ/nℤ: ⟨k⟩ = ⟨gcd(k, n)⟩.
Complex numbers
a + bi = re^{iθ}, r = √(a² + b²).
Moduli multiply, angles add.
DeMoivre: (re^{iθ})ⁿ = rⁿe^{inθ}.
Cyclic subgroup ⟨a⟩
⟨a⟩ = {aᵏ : k ∈ ℤ}, negative k too.
Smallest subgroup containing a.
G cyclic ⟺ G = ⟨a⟩; a a generator.
|a| = |⟨a⟩|.
Generators
aᵏ generates ⟺ gcd(k, n) = 1.
φ(n) generators; in ℤ/nℤ: [k].
Cyclic (ℤ/mℤ)ˣ: φ(φ(m)) generators.
φ(p^e) = p^e − p^{e−1}, multiplicative.
Coset criteria
g₁H = g₂H ⟺ Hg₁⁻¹ = Hg₂⁻¹
⟺ g₁H ⊆ g₂H ⟺ g₂ ∈ g₁H
⟺ g₁⁻¹g₂ ∈ H. So gH = H ⟺ g ∈ H.
counterexample
Cosets
gH = {gh}, Hg = {hg}; in ℤ/nℤ: a + H.
Cosets partition G; |gH| = |H|.
[G : H] = #left = #right cosets.
Cosets in examples
List: take uncovered g, form gH, repeat.
In Sₙ, gH: h acts first, then g.
S₄ by ⟨(1234)⟩, V, ⟨(123)⟩.
S₃ ≤ S₄: left ≠ right cosets.
Converse fails
d | |G| does not force
a subgroup of order d:
A₄ has none of order 6.
Lagrange's Theorem
H ≤ G finite ⇒ |H| divides |G|.
|G| = [G : H]·|H|.
[G : K] = [G : H][H : K].
Corollaries
|g| divides |G|; g^{|G|} = e.
|G| = p ⇒ cyclic, any g ≠ e generates.
gcd(|H|, |K|) = 1 ⇒ H ∩ K = {e}.
Euler and Fermat
gcd(a, n) = 1 ⇒ a^{φ(n)} ≡ 1 (mod n).
p ∤ a ⇒ a^{p−1} ≡ 1 (mod p); bᵖ ≡ b.
Reduce exponent mod φ(n), not n.
Repeated squaring for aᵏ mod n.
Normal subgroups
H ⊴ G: ghg⁻¹ ∈ H for all g, h.
⇒ gH = Hg for all g.
Abelian or index 2 ⇒ normal.
gH = Hg does not force gh = hg.
Counting arguments
Cosets partition G: sum their sizes.
Order-p elements come in
groups of p − 1.
|G| even ⇒ some g ≠ e has g = g⁻¹.
Prove or disprove
Counterexamples: (ℤ/8ℤ)ˣ,
ℤ/2ℤ × ℤ/2ℤ, S₃, ℚ, A₄.
Salvage: add the missing hypothesis.
Proving structure
Exhibit a generator ⇒ cyclic.
Every g² = e ⇒ abelian.
No proper nontrivial subgroups
⇒ cyclic of prime order.
Subgroup tests
H ≠ ∅ and ab⁻¹ ∈ H ⇒ H ≤ G.
Finite H: closure is enough.
Intersections of subgroups
are subgroups.
Direct products
|(g, h)| = lcm(|g|, |h|).
ℤ/mℤ × ℤ/nℤ ≅ ℤ/mnℤ ⟺ gcd(m, n) = 1
Internal: G = HK, H ∩ K = {e},
hk = kh ⇒ G ≅ H × K.
Unit groups
(ℤ/nℤ)ˣ: units mod n, order φ(n).
Identify: list unit orders, match
against products of ℤ/dℤ.
(ℤ/8ℤ)ˣ ≅ (ℤ/12ℤ)ˣ; (ℤ/24ℤ)ˣ ≅ (ℤ/2ℤ)³.
Isomorphism
Bijection, φ(ab) = φ(a)φ(b).
φ(e) = e, φ(a⁻¹) = φ(a)⁻¹, |φ(a)| = |a|.
≅ is an equivalence relation.
|G| = p
Cayley's Theorem
Every group ≅ a group of
permutations, via g ↦ (x ↦ gx).
Invariants (prove ≇)
Order, abelian, cyclic,
elements of each order, center.
D₈ ≇ Q₈, S₃ ≇ ℤ/6ℤ,
ℤ/4ℤ ≇ ℤ/2ℤ × ℤ/2ℤ.
Classification
Infinite cyclic ≅ ℤ; order n ≅ ℤ/nℤ.
|G| = p ⇒ G ≅ ℤ/pℤ.
Order 6: ℤ/6ℤ or S₃.


Input
x1
x2
x3
Hidden 1
h1
h2
h3
h4
Hidden 2
g1
g2
g3
g4
cat
dog


Continuous integration
Repository
Commit
signed, linked to an issue
Lint and format
style is not a review topic
Unit tests
fast, run on every push
Build
reproducible, pinned toolchain
Integration tests
real database and queue
pull request
Code review
one approval, two for core
changes requested
Rollback
one command, rehearsed monthly
Release gate
all checks green
changelog entry present
errors up
Delivery
Monitoring
error rate, latency, saturation
Production
promote
Canary
5% of traffic for an hour
Staging deploy
production-shaped data
Artifact store


Regular languages
week 1
Lexers
week 2
Grammars
week 3
LL parsing
project 1 due
week 4
LR parsing
week 5
Syntax trees
week 6
Midterm
week 7
Name resolution
project 2 due
week 8
Break
week 9
Type checking
week 10
IR design
week 11
Optimization
project 3 due
week 12
Code generation
week 13
Final exam
week 14


Web and mobile clients
API gateway
auth, rate limits, routing
Users
Cart
Orders
Payments
Inventory
reserve
charge
Postgres
Redis
Postgres
Ledger
Postgres